Metaheuristic Algorithms and Applications: Program Report I

Deadline: April 20, 2026

Please apply at least three of the following algorithms, GA, PSO, GSA, DNA, AIM, ABC, Queen-Bee GA or any other heuristic algorithms to determine the global minimum values $\boldsymbol{x}^*$ and the corresponding fitness value F^* of the following benchmark functions and make comparisons. Where the initial population size is 50, the maximum iteration cycle (generation) is 100 or 1000, and a total of 50 runs.

Unimodal benchmark functions.

Benchmark function	Dim.	Search Range	Optima
Schwefel 2.22 $F_2(x)=\sum_{i=1}^{D}\lvert x_i\rvert+\prod_{i=1}^{D}\lvert x_i\rvert$	2	$[-10, 10]^2$	0
Step $F_6(x)=\sum_{i=1}^{D}([x_i+0.5])^2$	2	$[-100, 100]^2$	0

Multimodal benchmark functions with fixed dimension.

Benchmark function	Dim.	Search Range	Optima
Rastrigin $F_9(X)=\sum_{i=1}^{n}\left[x_i^2-10\cos(2\pi\cdot x_i)+10\right]$	2	$[-5.12, 5.12]^2$	0
Griewank $F_{11}(x)=\frac{1}{4000}\sum_{i=1}^{D}x_i^2-\prod_{i}^{D}\cos(\frac{x_i}{\sqrt{i}})+1$	2	$[-600, 600]^2$	0
Penalized $F_{13}(x)=0.1\{\sin^2(\pi 3x_1)+\sum_{i=1}^{D-1}(x_i-1)^2[1+\sin^2(3\pi x_{i+1})]+(x_D-1)^2[1+\sin^2(2\pi x_D)]\}+\sum_{i=1}^{D}u(x_i,5,100,4)$ $u(x_i,a,k,m)=\begin{cases}k(x_i-a)^m & x_i>a\\ 0 & -a\le x_i\le a\\ k(-x_i-a)^m & x_i<-a\end{cases}$	2	$[-50, 50]^2$	0

Benchmark function	Dim.	Search Range	Optima
Kowalik $F_{15}(x)=\sum_{i=1}^{11}[a_i-\frac{x_1(b_i^2+b_ix_2)}{b_i^2+b_ix_3+x_4}]^2$	4	$[-5,5]^4$	0.0003075

Benchmark function	Dim.	Search Range	Optima
Branin $F_{17}(x) = (x_2 - \frac{5.1}{4\pi^2}x_1^2 + \frac{5}{\pi}x_1 - 6)^2 + 10(1 - \frac{1}{8\pi})\cos x_1 + 10$	2	$[-5,10]\times[0,15]$	0.398

Table A.2
a_i and b_i in F_{15}.

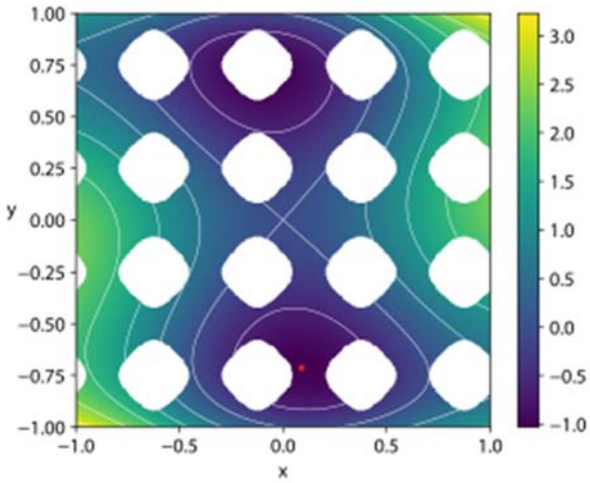
i	1	2	3	4	5	6	7	8	9	10	11
a_i	0.1957	0.1947	0.1735	0.1600	0.0844	0.0627	0.0456	0.0342	0.0342	0.0235	0.0246
b_i^{-1}	0.25	0.5	1	2	4	6	8	10	12	14	16

This program report should include "programs" and "discussions". For discussion part, the following results should be investigated.

1. Plot the 3D plot of these functions with D=2.
2. Plot the searching results of one of the 50 runs of these functions.
3. Plot the progress of the average best-so-far solution vs. iteration over 50 runs with 100 and 1000 iterations for these functions and make some comments or interpretations.
4. List a table of the averaged results over 50 runs with 100 and 1000 iterations for solutions of average best-so-far, average mean fitness, and the median best-so-far solution in the last iteration.
5. List a table of the averaged results (including mean and standard deviation) over 50 runs with 100 and 1000 iterations for the best solution in the last iteration.
6. **Give the mean and standard deviation of the distance errors of the best-so-far particle with the desired optima, $\|x_{best} - x^*\|$.**
7. Please also give all parameters setting in your simulations.

Extra --- Test functions for multi-objective optimization

Gomez and Levy function (modified)



$$f(x,y) = 4x^2 - 2.1x^4 + \frac{1}{3}x^6 + xy - 4y^2 + 4y^4$$

Subjected to :

$$-\sin(4\pi x) + 2\sin^2(2\pi y) \leq 1.5$$

$$-1 \leq x \leq 0.75, \quad -1 \leq y \leq 1$$

$$f(0.08984201, -0.7126564) = -1.031628453$$